

Early Risk Detection for Student Success

OULAD Case Study · Stakeholder Report

1. Executive Summary

The Data Science Academy's Student Success Office needs to identify students at risk of failing or withdrawing early enough to act, and to understand *why* a student was flagged so advisors can offer the right kind of support. This report summarizes an end-to-end analysis of the Open University Learning Analytics Dataset (OULAD), covering 32,593 student-course records, built to answer that need.

What We Built

- Cleaned and integrated seven raw OULAD tables into a single student-course record.
- Engineered an early-warning feature set using only information available within the first 28 days of a course.
- Compared a Logistic Regression baseline against a Random Forest ensemble, validated with student-grouped cross-validation.
- Explained model predictions both globally (what drives risk overall) and individually (why one student was flagged).
- Tested how early a reliable signal appears, from Day 7 through Day 28, and identified the single strongest early assessment checkpoint.

Headline Results

52.8%	0.872	Day 7	95.7%
of records are At Risk (Fail/Withdrawn)	Random Forest Average Precision	earliest point with a meaningful signal	at-risk rate with zero early platform activity

Bottom line

A student's early engagement with the course website is the strongest operational warning signal available, and it can be generated within the first week of a course, well before any exam result is known.

This document gives a concise, non-technical summary for stakeholders. The complete analysis, including all code, statistical checks, and intermediate results, is available in the accompanying GitHub repository (8 Jupyter notebooks, each documenting one stage of the pipeline).

2. Data & Methodology

Dataset

The Open University Learning Analytics Dataset (OULAD) combines seven tables: student demographics, module registration and withdrawal dates, assessment results, course-website (VLE) activity logs, and course metadata. After cleaning and merging, the analysis works at the level of one row per student-course enrollment (32,593 rows).

Data Cleaning Highlights

- Removed 787,170 exact-duplicate rows from the 10.6M-row VLE activity log.
- Verified referential integrity across all tables (every student ID, assessment ID, and site ID resolves correctly) before merging.
- Missing values were treated as information where appropriate, rather than deleted. For example, a missing withdrawal date corresponds to a student who did not withdraw in 99.9% of cases.
- Added explicit missingness indicators (e.g. `no_early_vle_activity`) instead of silently imputing zeros, since the absence of early activity is itself a meaningful signal.

Defining "At Risk"

The case study asks for students at risk of *failing or withdrawing*. We therefore define:

- **At Risk** = Fail or Withdrawn (52.8% of records)
- **Not At Risk** = Pass or Distinction (47.2% of records)

This split is close to balanced, which is favorable for classification: standard metrics remain reliable without needing aggressive resampling techniques.

Early-Warning Design

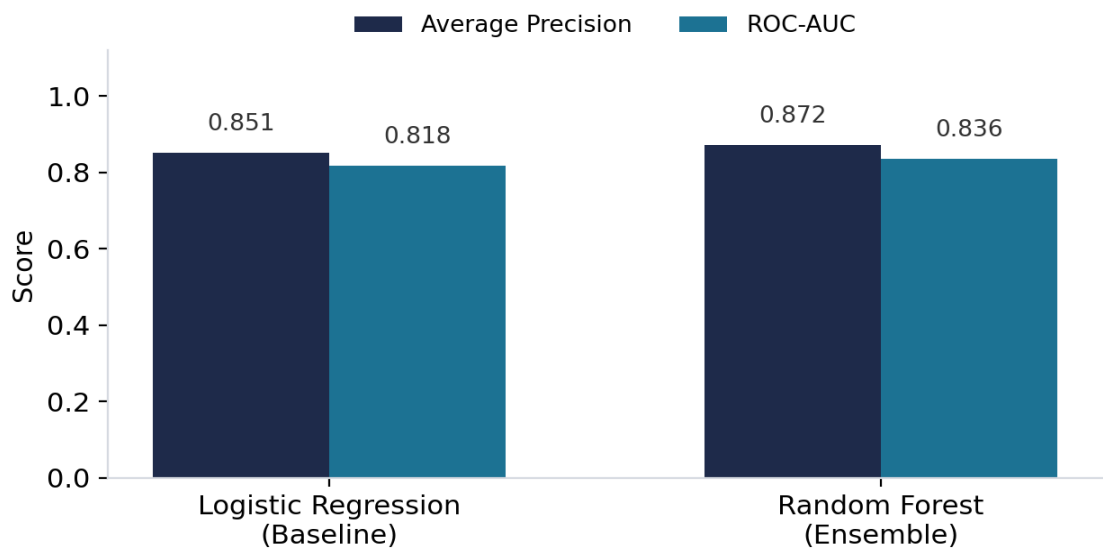
All predictive features are restricted to information available on or before **Day 28** of a course: assessment scores, submission timing, and VLE activity from that window only. Any column that could only be known after the course concluded (including the exact withdrawal date) was explicitly excluded to prevent the model from "cheating" with future information.

Models & Validation

Two models were trained and compared on identical inputs: a **Logistic Regression** baseline (simple, fast, directly interpretable) and a **Random Forest** ensemble (expected to capture non-linear patterns and feature interactions). Validation used **Stratified Group K-Fold** cross-validation, grouping by student ID so that no student who appears in more than one course record can leak information between training and evaluation.

3. Model Results

Model Comparison: Baseline vs Ensemble



Metric	Logistic Regression (Baseline)	Random Forest (Ensemble)
Average Precision	0.8512	0.8716
ROC-AUC	0.8179	0.8358
Precision	0.7634	0.8005
Recall	0.7243	0.7108
F1 Score	0.7433	0.7529
Accuracy	0.7358	0.7538

Why Random Forest Was Selected

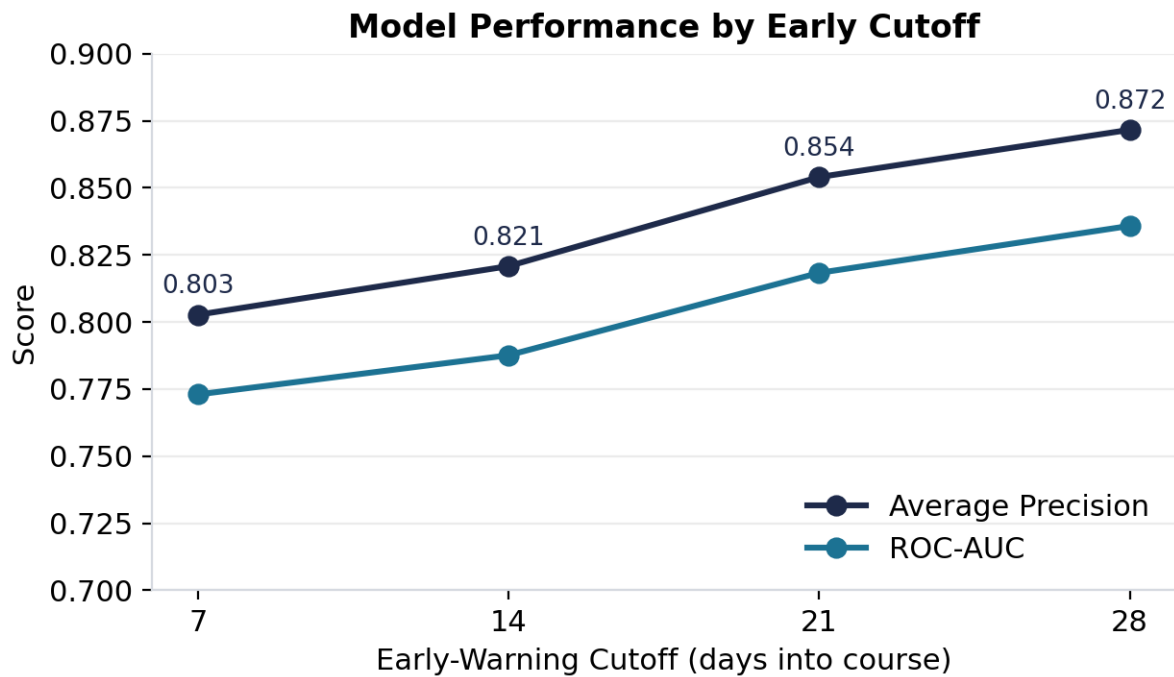
Random Forest outperforms the baseline on every metric except recall, where the two are close (0.711 vs 0.724). The largest practical gain is in **Precision** (0.80 vs 0.76): when Random Forest flags a student as at risk, it is right more often, which matters directly for an advising team with limited capacity. Fewer advisor hours are spent investigating students who turn out not to need help.

Confusion Matrix (Random Forest, out-of-fold predictions)

	Predicted: Not At Risk	Predicted: At Risk
Actual: Not At Risk	12,337	3,048
Actual: At Risk	4,976	12,232

Of 17,208 genuinely at-risk records, the model correctly identifies 12,232 (71.1%) and misses 4,976 (28.9%). This trade-off, and the operating threshold used to produce it, can be adjusted based on the Student Success Office's actual advising capacity.

4. Early Warning: How Soon Can We Know?



The same modeling approach was tested using progressively larger early-data windows: Day 7, 14, 21, and 28. Performance improves steadily as more information accumulates, but a **usable signal is already present after just one week**, before a single early assessment has typically even taken place.

Cutoff	Average Precision	ROC-AUC
Day 7	0.8027	0.7730
Day 14	0.8208	0.7876
Day 21	0.8540	0.8183
Day 28	0.8716	0.8358

Strongest Early Assessment Checkpoint

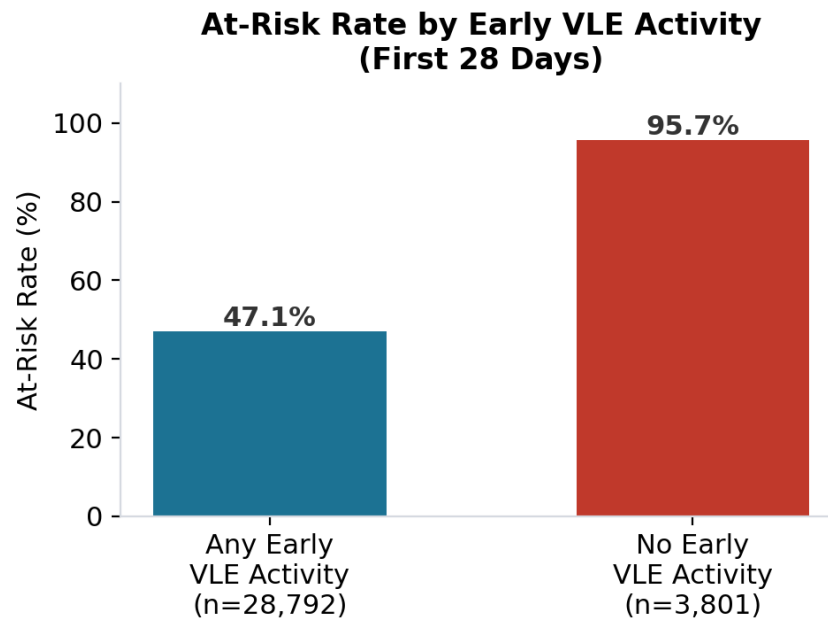
Among all individual assessments scheduled within the first 28 days, one stands out: the **CCC module, 2014B presentation, CMA assessment on Day 18**. Among 1,346 students with a recorded score, not-at-risk students averaged **74.5** while at-risk students averaged **47.5**, a 27-point gap and the single strongest individual academic checkpoint found in this analysis (risk-signal AUC of 0.790).

Operational takeaway

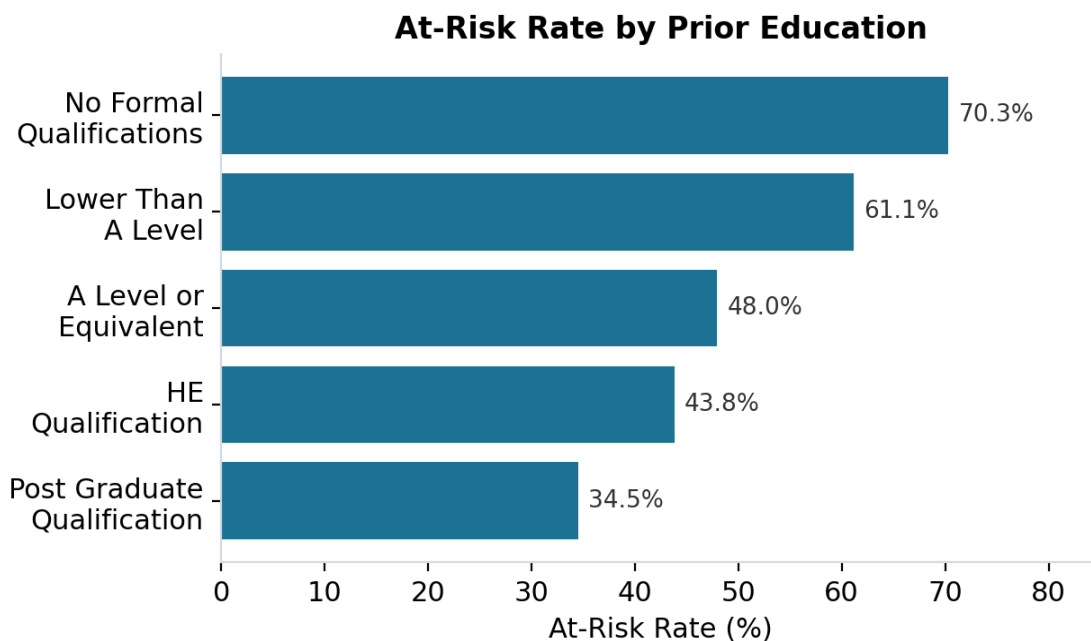
The Student Success Office does not need to wait until Day 28 for a first alert. A staged workflow is recommended: screen from Day 7, refine the risk list as Day 14 and 21 data arrive, and treat performance on early assessments like the Day-18 CCC checkpoint as an additional, targeted trigger for follow-up.

5. What Drives Risk

Two complementary views were used to understand what separates at-risk from not-at-risk students: patterns observed directly in the data (exploratory analysis), and what the trained model actually relies on (permutation importance). Both point to the same conclusion: **engagement matters more than any single background factor**.



Students with zero recorded activity on the course website in the first 28 days have a **95.7% at-risk rate**, versus 47.1% for students with any activity at all, roughly a 2x gap and the widest observed behavioral gap identified anywhere in this analysis. (Note: as a formal model feature, `course_module` ranks highest by permutation importance; engagement is the strongest overall feature group.)



Other Contributing Factors

- **Socioeconomic background:** risk falls from 64.8% in the most deprived areas to 42.5% in the least deprived.
- **Age:** risk falls from 55.0% for students under 35 to 38.4% for students 55 and older.
- **Course module:** risk ranges from 29.0% (module AAA) to 62.2% (module CCC), showing that course context matters independently of the student.
- **Engagement intensity:** at-risk students show roughly half the activity of not-at-risk students across every measure, including median clicks (141 vs 292), active days (9 vs 16), and distinct sites visited (19 vs 27).
- **Top global model feature (permutation importance):** course_module, confirming that course context is the single most influential input to the model overall.

These are associations observed in the data, not proof that any one factor directly causes poor outcomes. They should guide where to look and how to support students, not be used as a deterministic or punitive rule.

6. Business Questions & Recommendations

Q1. Which factors most reliably predict risk?

A combination of course context, early engagement, prior education, and socioeconomic background. Engagement (VLE activity) is the strongest overall feature group, while course_module is the single highest-ranked individual feature by permutation importance.

Q2. How early can we flag an at-risk student?

A meaningful signal is present from Day 7 (AP 0.803), improving steadily through Day 28 (AP 0.872).

Q3. Do demographic groups face different risk levels?

Yes. Prior education shows the widest gap (70.3% to 34.5%), followed by socioeconomic band (64.8% to 42.5%) and age (55.0% to 38.4%).

Q4. How does website activity relate to outcome?

Strongly. At-risk students show roughly half the engagement of not-at-risk students on every measure, and having zero early activity at all nearly doubles the risk rate (95.7% vs 47.1%).

Q5. Which assessments carry the strongest early signal?

The CCC / 2014B CMA assessment on Day 18, with an 27-point score gap between at-risk and not-at-risk students.

Q6. How can advisors use a flagged prediction?

Each flagged student comes with the specific factors driving that prediction (e.g. low early engagement, weak early assessment performance), letting an advisor choose a targeted response, such as an engagement nudge versus academic tutoring, rather than a generic warning.

Recommended Advisor Workflow

Early data → Risk score → Explanation → Advisor verification → Targeted support → Risk re-scored as new data arrives

The output of this system is not simply a list of "high-risk" students. It is a continuously updated, explained risk score that an advisor reviews before deciding on a course of action, keeping a human in the loop for every intervention decision.

7. Limitations & Conclusion

Limitations

- The Day-28 early-warning cutoff is a project design choice, not a fixed system requirement; it can be adjusted.
- The target is based on final Fail/Withdrawn outcomes, so this is a retrospective predictive study rather than a live, deployed intervention system.
- Feature importance describes model behavior and observed associations, not proven causal effects.
- Individual explanations vary one feature at a time and do not fully capture interactions between features.
- The model was trained and evaluated on OULAD; it should be re-validated before use with a different student population or institution.
- The operating threshold for flagging students should be chosen based on actual advising capacity and the relative cost of a missed student versus an unnecessary alert.

Conclusion

This project demonstrates a practical early-risk workflow using the OULAD dataset. A Random Forest model, validated with student-grouped cross-validation, outperforms a simple baseline (0.872 vs 0.851 Average Precision) and produces a meaningful warning signal as early as the first week of a course. The strongest single driver of risk is early engagement with the course platform, and the strongest individual academic checkpoint is the Day-18 CCC assessment. Together, these findings support a staged, explainable early-warning system that directs limited advising resources toward the students who need them most, well before a missed assignment or failed exam makes the risk obvious.

Full technical documentation: All code, statistical validation, and additional detail are available in the project's GitHub repository, organized as 8 sequential Jupyter notebooks covering data understanding, cleaning, feature engineering, exploratory analysis, modeling, interpretability, early-warning timing, and this stakeholder report.

github.com/SmurFxD-ReallOne/Early-Risk-Detection-OULAD